

Name: _____

Section: _____

Std ID: _____

National University of Computer & Emerging Sciences- FAST

Karachi Main Campus

Department of Computer Science

Final Examination-Fall 2009

MT101- Calculus I

Dated: 9 December 2009

Max Marks: 100

Max Time: 3 hours

Attempt all questions:**Show all necessary working and steps:**

Q.1 Solve the inequality and sketch the solution on a coordinate line. (20)

a) $\frac{2x-5}{x-2} < 1$

b) Solve for x express the solution $\frac{3}{|2x-1|} \geq 4$

c) Sketch the graph of $y = 2 + \sqrt{9-x^2}$

Q.2 Find the first derivative of the functions (25)

i) $y = \sqrt{1 + \sin^3(xy^2)}$ (ii) $3xy = (x^3 + y^2)^{\frac{3}{2}}$ (iii) $y = \tan^{-1} \sqrt{\frac{1-x}{1+x}}$

(iv) $y = \sec^{-1}(x^7 + 1)$ (v) $y = (\ln x)^{\tan x}$

Q.3 a) Prove that $\tanh^{-1} x = \frac{1}{2} \ln \frac{1+x}{1-x}$ (5)

b) Prove that $\int_0^{\frac{\pi}{2}} \cos^n x dx = \frac{2.4.6 \dots (n-1)}{3.5.7 \dots n}$ (n is odd integer) (5)

c) Find the exact value of the given quantity. $\sin \left[\sin^{-1} \left(\frac{2}{3} \right) + \cos^{-1} \left(\frac{1}{3} \right) \right]$ (5)

Q.4 Express $1 + 2 + 2^2 + 2^3 + 2^4 + 2^5$ in sigma notation (5)a) $j=0$ as the lower limit of summation.b) $j=1$ as the lower limit of summation.c) $j=2$ as the lower limit of summation.

Q.5 Find the area under the curve of the function $y = 9 - x^2$ over the interval $[0, 3]$, where $n=6$.

By using right end point, mid point methods. Also comment which is the most suitable method. (7)

Q.6 Evaluate the integrals

(20)

i) $\int \frac{x^2 + x - 2}{3x^3 - x^2 + 3x - 1} dx$ (ii) $\int_1^6 \frac{dx}{x^4 \sqrt{x^2 + 3}}$ (iii) $\int_0^{\frac{\pi}{2}} \tan^{-1} \frac{x}{2} dx$ (iv) $\int \frac{dx}{\sqrt{x} + \sqrt[3]{x}}$

Q.7 Find limits

(8)

i) $\lim_{x \rightarrow \infty} \frac{x}{x^2 - 4}$ ii) $\lim_{x \rightarrow -\infty} \frac{x - 2}{x^2 + 2x + 1}$